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ML Project: AI-Based Sentiment Classification for Customer Reviews

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Abstract

This paper presents a machine-learning solution for automating sentiment classification of e-commerce customer reviews, addressing the inefficiency of manual analysis and the limitations of existing systems. The study investigates key challenges such as contextual ambiguity, sarcasm, linguistic variability, and extreme class imbalance in large-scale review datasets. To overcome these issues, a Stacking Ensemble Classifier was developed, combining Multinomial Naive Bayes, Random Forest, and Linear Support Vector Classifier models, with Logistic Regression serving as the meta-learner. Comprehensive preprocessing steps—including text cleaning, lemmatization, and TF-IDF vectorization—were applied to enhance data quality, while Random Oversampling was used to mitigate class imbalance.

Experimental results demonstrate that the ensemble model achieves a validation accuracy of **0.8263**, showing strong performance in detecting positive and negative sentiment, with recalls of **95%** and **88%**, respectively. However, neutral sentiment remains challenging, achieving only **26%** recall, indicating a need for more advanced context-aware methods. Overall, the proposed model provides a scalable, reliable framework for real-time sentiment analysis and offers a foundation for future enhancements such as Transformer-based architectures.

Keywords

Machine Learning, Sentiment Classification, Stacking Ensemble, Text Mining, Customer Reviews, Natural Language Processing (NLP)

1. Introduction

The exponential rise of online platforms and e-commerce has resulted in a massive amount of customer reviews, rendering manual analysis inefficient and unfeasible. Using machine learning for sentiment categorization provides a scalable and automated approach to getting useful information from this unstructured textual material. This study addresses the urgent need for reliable and effective sentiment analysis methods that can account for the nuances and intricacies of human language in consumer input. Our goal is to develop a cutting-edge ML module that addresses existing constraints and delivers actionable business intelligence, which will ultimately improve customer experience and promote business expansion in
online marketplaces.

2. Problem Definition

2.1 Current System Description

E-commerce platforms and review aggregators handle thousands of customer reviews daily. Current analysis methods range from manual reading to basic automated systems that often fail to capture contextual nuances, sarcasm, and domain-specific language patterns.

2.2 Challenges and Inefficiencies

- Manual review analysis is expensive and takes a lot of time
- With slang and casual language, current automated systems demonstrate minimal accuracy.
- Bad performance with a combination of emotions and sarcastic remarks
- Restricted ability to adapt to diverse linguistic situations and particular commercial industries

2.3 Motivation for ML Integration

Machine learning integration can:

- Offer real-time mood analysis at scale

- Enhance the accuracy of understanding customer preferences

- Facilitate continuous product improvements and proactive customer care

- Improve the entire client experience and business insights

3. Literature Review

3.1 Current State of the Art

Recent developments in customer review sentiment analysis show a definite shift from classic machine-learning methods to deep learning and Transformer-based models. Because of their simplicity, interpretability, and inexpensive processing cost, conventional machine-learning classifiers like Support Vector Machines (SVM), Logistic Regression, Naive Bayes, K-Nearest Neighbors, Decision Trees, and Random Forest have been widely utilized. These methods, which frequently employ n-gram, Bag-of-Words, or TF-IDF features, have fair performance, with some research indicating accuracies in the range of 85% to 92%. However, their main drawback is their failure to comprehend textual context and long-term relationships.

Notably, LSTM and hybrid CNN-LSTM models, both of which use deep learning, have made notable advancements in recognizing sequential patterns in customer reviews. The hybrid CNN-LSTM approach achieved accuracies exceeding 90% in a number of studies, with the highest accuracy reaching 94% because of its fusion of feature extraction and memory capabilities. Even with this advancement, these models still have trouble with more complex contextual connections and need huge datasets to function well.

Transformer-based models, notably BERT and ELECTRA, currently stand out as the best in the field. Their greater capacity for comprehending contextual semantics and managing multilingual or domain-specific datasets was demonstrated by a number of researches. ELECTRA attained 98% accuracy and the highest F1-score (0.971) among all the models assessed, while BERT had accuracies that varied from 89% to 95%. Despite their outstanding performance, these models need a lot of processing power and a lot of training.

3.2 Comparative Analysis

Model Type	Best Reported Accuracy	Strengths	Limitations
SVM	97%	Strong baseline; effective for short texts	Limited contextual understanding
Logistic Regression	90%+	Stable and interpretable	Struggles with complex linguistic structures
Naïve Bayes	74–87%	Fast and lightweight	Poor with contextual and neutral sentiment
Random Forest	90%	Good interpretability	High cost on larger datasets
LSTM	87–96%	Learns long-term dependencies	Requires large datasets
CNN-LSTM	94%	Strong hybrid feature representation	Does not capture deep contextual meaning
BERT	89–95%	Excellent contextual understanding	High computational demand
ELECTRA	98%	Best overall performance	Long runtime and resource-intensive

Table 1 Performance comparison of sentiment models

Overall, Transformer models consistently outperform both classical and deep learning approaches, especially in multilingual and domain-specific settings such as e-commerce.

A comparative evaluation of traditional machine-learning models, deep-learning architectures, and Transformer-based approaches reveals clear performance distinctions across the reviewed studies. Traditional models such as SVM, Logistic Regression, and Naive Bayes demonstrate decent accuracy levels, often ranging between 85% and 92%. Their strengths lie in their interpretability, computational efficiency, and ability to perform well on smaller datasets using features such as TF-IDF and n-grams. However, these models struggle to capture contextual meaning, sarcasm, long-range dependencies, and subtle sentiment cues, limiting their effectiveness for real-world e-commerce reviews. Deep-learning models, particularly LSTM and hybrid CNN-LSTM architectures, show noticeable improvement by learning sequential patterns and extracting richer features. These models often exceed 90% accuracy, as shown in Table 1, but still require large amounts of labeled data and lack the ability to fully model complex linguistic structures.

In contrast, Transformer-based models such as BERT and ELECTRA consistently outperform both traditional and deep-learning methods because of their superior contextual understanding and bidirectional encoding of sentences. ELECTRA achieved the highest reported accuracy and F1-score among the compared models, reflecting its ability to handle domain-specific vocabulary, multilingual inputs, and ambiguous sentiment expressions more effectively. Despite their computational cost, Transformers offer the best balance between classification accuracy and real-world applicability, particularly for e-commerce platforms dealing with diverse and high-volume customer feedback. These findings highlight a clear progression in sentiment-analysis research from simple statistical models to advanced architectures capable of capturing nuanced language representations.

3.3 Gap Analysis

Despite the progress in sentiment analysis research, several gaps remain evident across the reviewed studies:

1) Weak performance on neutral sentiment:

Although the majority of models accurately categorize positive and negative emotions, they have difficulty with neutral reviews, which are prevalent in e-commerce data.

2) Lack of aspect-based sentiment analysis (ABSA):

The majority of research merely categorizes general mood without mentioning the particular elements that are being evaluated (e.g., product quality, delivery, price).

3) Limited research on Arabic sentiment analysis:

Due to the concentration of most models and datasets on English or Chinese, there is a void in Arabic-language e-commerce applications.

4) Data imbalance issues:

Uneven distributions of positive, negative, and neutral reviews were reported in a number of research, which had a significant impact on the model's performance, particularly for minority groups.

5) High computational cost of Transformer models:

Models such as BERT and ELECTRA provide cutting-edge performance but need a lot of training time and processing power, which makes real-time deployment difficult.

6) Reliance on translation tools:

The reliability of the models was diminished because certain bilingual datasets included semantic loss as a result of machine translation.

3.4 Justification for the Proposed Study

Based on the identified gaps, the proposed sentiment-classification module aims to address these limitations by:

- Making use of a Transformer-based model that is able to understand the nuances of context in customer reviews in both English and Arabic.
- By using superior preprocessing and balanced datasets, the categorization of neutral sentiment can be improved.
- Creating a system that may be immediately integrated into e-commerce platforms to aid in real-time business choices.
- Improving the dataset's representativeness and balance, particularly for multi-class sentiment analysis.
- Using a more domain-specific model that is focused on analyzing client feedback as opposed to analyzing text in general.

4. ML Module Proposal

4.1 Objectives

The primary goal of this ML module is to perform sentiment classification on customer reviews by categorizing them into three classes: positive, neutral, and negative. This module improves the system by enabling automated feedback analysis, early detection of dissatisfaction, and clear insights into user behavior. It replaces manual review with a scalable, consistent, and data-driven process that supports better decision-making across the system.

4.2 Approach: Stacking Ensemble Classifier

The proposed solution uses a Stacking Ensemble model to maximize predictive accuracy by combining the strengths of multiple classifiers:

- Base Models:
  - Multinomial Naive Bayes (MNB): Chosen because it is effective when the number of features is very large and efficient at identifying common word patterns.
  - Linear Support Vector Classifier (Linear SVC): Chosen because it performs very well on high-dimensional TF-IDF features and is capable of drawing clear decision boundaries.
  - Random Forest: Chosen because it performs well on high-dimensional TF-IDF features and is relatively robust to noise, capturing non-linear patterns.

- Preprocessing

  All text reviews went through a thorough and consistent preparation pipeline:
  - We first converted the raw numerical review ratings into our three target sentiment categories.
  - The text was cleaned by removing URLs, converting to lowercase, stripping punctuation, filtering out common stopwords, and applying lemmatization (reducing words to their root form).
  - The cleaned text was then transformed into numerical features using a TF-IDF Vectorizer (set to 50,000 features with a bi-gram range of 1,2).
- Class Imbalance Handling: The raw dataset was heavily skewed toward positive reviews. To prevent the model from ignoring the critical but rare negative and neutral classes, we corrected this imbalance by applying Random Oversampling specifically to the training data. This ensures all sentiment categories are represented during the learning phase.

- Stacking Mechanism: The base models generate probability predictions for all sentiment classes. These probabilities are combined into new features, which the Logistic Regression meta-classifier (C = 0.1) uses to make the final predictions. The meta-model learns from the strengths and weaknesses of each base model to improve overall performance.

4.2.1 Detailed Implementation of the ML Module

- Data preprocessing

  In this stage of the project, I performed the data preprocessing and preparation required for the sentiment-classification models. The process began by uploading and loading the dataset (sample.json) using Pandas, followed by converting the numerical review ratings into three sentiment categories: positive, neutral, and negative. I then applied several text-cleaning steps, including removing URLs, converting text to lowercase, removing punctuation, eliminating stopwords, and applying lemmatization to reduce words to their base form. These steps help improve the quality and consistency of the textual data.

  After preprocessing the text, the dataset was split into three sets: training (70%), validation (15%), and testing (15%), while maintaining class balance using stratified splitting. Finally, the cleaned text was transformed into numerical features using a TF-IDF Vectorizer with 50,000 features and a bi-gram range (1,2). The resulting matrices (X_train, X_val, X_test) were prepared for use by the rest of the team to train and evaluate different machine-learning models.

- Random Forest Classifier

  For the first base model in our ensemble, I implemented a Random Forest classifier. This model was chosen because it performs well on high-dimensional TF-IDF features and is relatively robust to noise in text data. Since customer reviews vary widely in writing style and vocabulary, Random Forest serves as a reliable baseline before moving to more complex learners.

  The dataset suffered from severe class imbalance, with positive reviews dominating the majority of samples. To address this, I applied Random Oversampling on the training set, which expanded it from around 70,000 to about 160,000 samples. This step significantly improved the model's ability to detect minority classes, especially negative and neutral reviews.

After converting the text into TF-IDF vectors (50,000 features including unigrams and bigrams), I tuned the model using RandomizedSearchCV. The best parameters consisted of 100 trees, a maximum depth of 20, and a minimum samples split of 5. These parameters provided a strong balance between model complexity and overfitting control.

On the validation set, the Random Forest classifier achieved an accuracy of **0.7179** The detailed performance metrics are shown in Figure 1. It performed very strongly on positive reviews, but it struggled with the neutral class, despite this, the model showed good recall for the negative class, making it valuable for capturing critical feedback. The confusion matrix is presented in Figure 2.

```
--- Random Forest Validation Classification Report ---
              precision    recall  f1-score   support

    negative       0.44      0.68      0.54      1863
     neutral       0.27      0.41      0.33      1648
    positive       0.92      0.77      0.83     11489

    accuracy                           0.72     15000
   macro avg       0.54      0.62      0.57     15000
weighted avg       0.79      0.72      0.74     15000
```

Figure 1 Random Forest Validation Classification Report

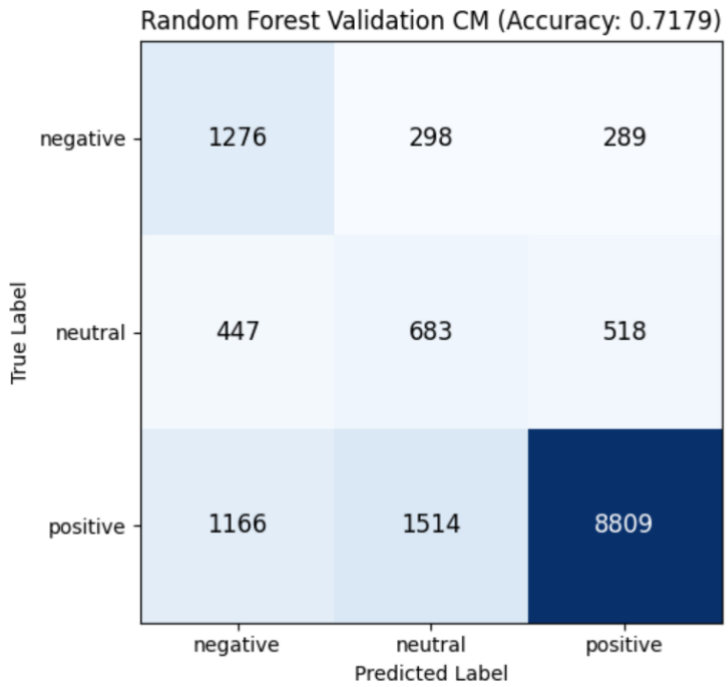

Figure 2 Random Forest Validation CM

- Multinomial Naive Bayes

  The Multinomial Naive Bayes classifier is one of the most widely used algorithms for text classification, particularly because it performs effectively when the number of features is very large, as is typical in text datasets. For this reason, we selected this model after converting the textual data into a TF-IDF representation.

  The model was then trained using Laplace smoothing (alpha = 1.0) to prevent zero-probability issues for rare words. Additionally, we addressed class imbalance by applying oversampling, ensuring that all sentiment classes were fairly represented during training. This step reduces bias toward the majority class and improves the model's overall classification performance.

  After training, the model showed strong performance on the validation set, achieving an accuracy of **0.73**. The results showed that the classifier performs very well on the positive class, with high precision and recall. However, it struggles to distinguish between the negative and neutral classes (especially the neutral class) which led to reduced F1-scores for these classes. The corresponding confusion matrix is shown in Figure 4.

  Overall, MultinomialNB is a strong and simple starting model for text sentiment classification. It learns common word patterns well, but it has difficulty handling rare cases or reviews with weak or unclear sentiment, especially neutral ones. The validation results are illustrated in Figure 3.

```
--- MultinomialNB Validation Classification Report ---
              precision    recall  f1-score   support

    negative       0.50      0.67      0.57      1863
     neutral       0.27      0.51      0.35      1648
    positive       0.94      0.77      0.85     11489

    accuracy                           0.73     15000
   macro avg       0.57      0.65      0.59     15000
weighted avg       0.81      0.73      0.76     15000
```

Figure 3 MultinomialNB Validation Classification Report

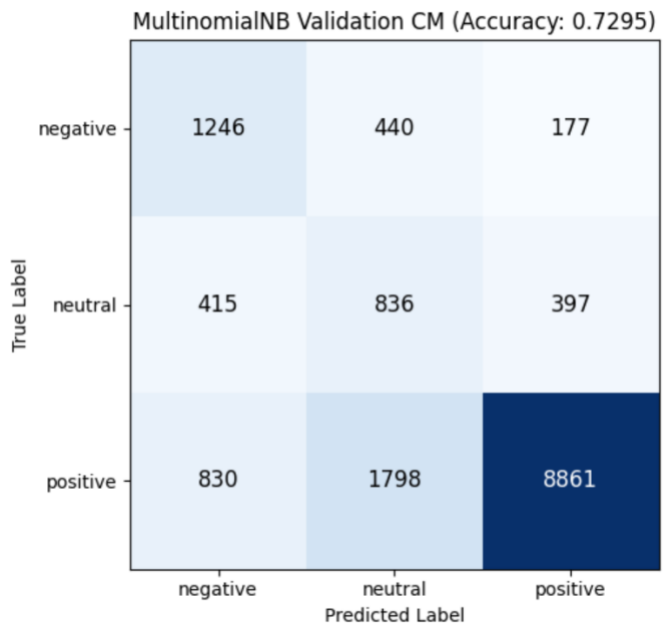

Figure 4 MultinomialNB Validation CM

- Linear Support Vector Classifier (Linear SVC)

  For the third base model in our ensemble, I implemented a Linear Support Vector Classifier (Linear SVC). This model is widely used in text classification because it performs very well on high-dimensional TF-IDF features and is capable of drawing clear decision boundaries between sentiment classes. Since customer reviews often contain subtle differences in wording, Linear SVC is particularly effective at separating closely related sentiment patterns.

  The model was tuned using RandomizedSearchCV, which selected hinge loss and a penalty parameter of C = 1. This configuration offered a strong balance between maximizing the margin and avoiding overfitting.

  On the validation set, Linear SVC achieved an accuracy of 0.8497, which is the highest performance among all the individual base models. The classification report is shown in Figure 5. The model demonstrated excellent performance in recognizing strong emotions, achieving a Recall of 90% for positive reviews and 88% for negative reviews. However, it struggled noticeably with the more nuanced neutral class, which achieved a lower Recall of 70%. This indicates that the model tends to misclassify ambiguous or mixed-tone reviews as either positive or negative. The confusion matrix for this model is presented in Figure 6.

```
--- LinearSVC Validation Classification Report ---
              precision    recall  f1-score   support

    negative       0.63      0.65      0.64      1863
     neutral       0.41      0.30      0.35      1648
    positive       0.90      0.93      0.91     11489

    accuracy                           0.82     15000
   macro avg       0.65      0.63      0.63     15000
weighted avg       0.81      0.82      0.82     15000
```

Figure 5 LinearSVC Validation Classification Report

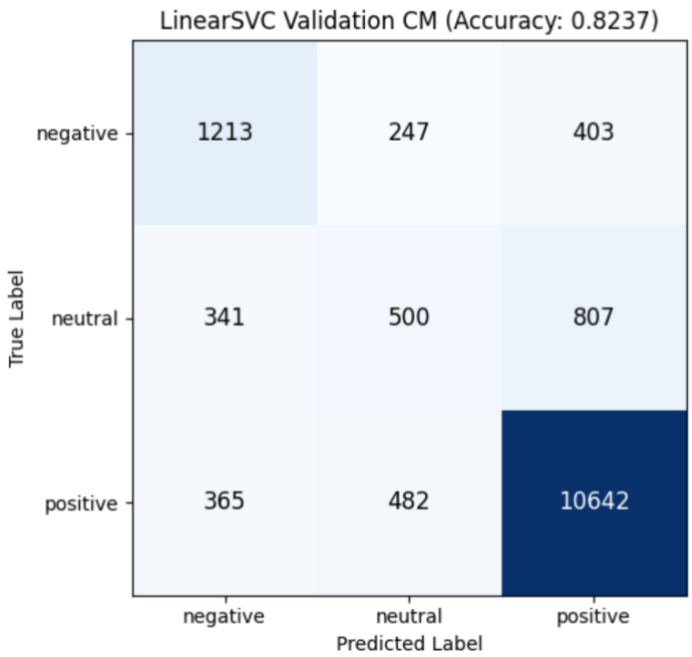

Figure 6 LinearSVC Validation CM

- Stacking Ensemble Meta-Learner

  To further improve sentiment classification, I implemented a stacking ensemble that combines the three base models Random Forest, Multinomial Naive Bayes, and Linear SVC. Each model brings complementary strengths, and the stacking approach allows the meta-model to learn how to combine their predictions effectively.

Base Models

- Random Forest: Pre-trained on TF-IDF features, capturing non-linear patterns.
- Multinomial Naive Bayes: Efficient at identifying common word patterns.
- Linear SVC: Calibrated with the sigmoid method to produce reliable class probabilities.

- Stacking Mechanism

  The base models generate probability predictions for all sentiment classes. These probabilities are combined into new features, which the Logistic Regression meta-classifier (C = 0.1) uses to make the final predictions. By learning from the strengths and weaknesses of each base model, the meta-model improves overall performance, particularly for challenging cases. The confusion matrix for the test set is illustrated in Figure 8.

- Performance

  The stacking ensemble achieved an accuracy of 0.8263 on the validation set. It excelled at detecting positive reviews (precision 0.88, recall 0.95) and performed reasonably on negative reviews (precision 0.68, recall 0.58). The neutral class remained challenging (precision 0.43, recall 0.26), with most errors occurring between neutral and negative reviews. Despite this, the ensemble provides a more balanced prediction across classes than most individual models, combining their strengths to improve overall robustness.

Overall, the stacking ensemble proves to be a powerful meta-learner. The final test classification results are shown in Figure 7. effectively integrating different base models to capture both strong and subtle sentiment signals. The ROC curves for all sentiment classes are plotted in Figure 9.

```
--- Stacking Ensemble TEST Classification Report ---
              precision    recall  f1-score   support

    negative       0.68      0.58      0.63      1863
     neutral       0.43      0.26      0.33      1649
    positive       0.88      0.95      0.91     11488

    accuracy                           0.83     15000
   macro avg       0.66      0.60      0.62     15000
weighted avg       0.80      0.83      0.81     15000
```

Figure 7 Stacking Ensemble TEST Classification Report

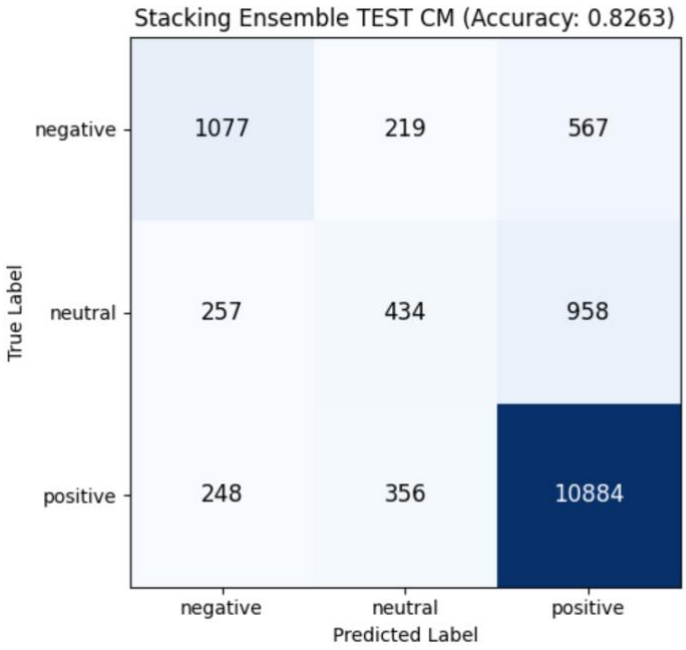

Figure 8 Stacking Ensemble TEST CM

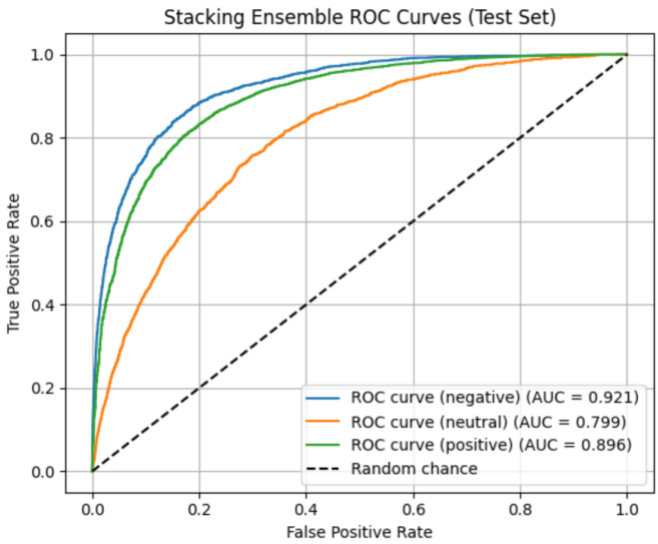

Figure 9 Stacking Ensemble ROC Curves (Test Set)

4.3 Workflow

The workflow of the sentiment classification module follows this sequence. The overall workflow is illustrated in Figure 10:

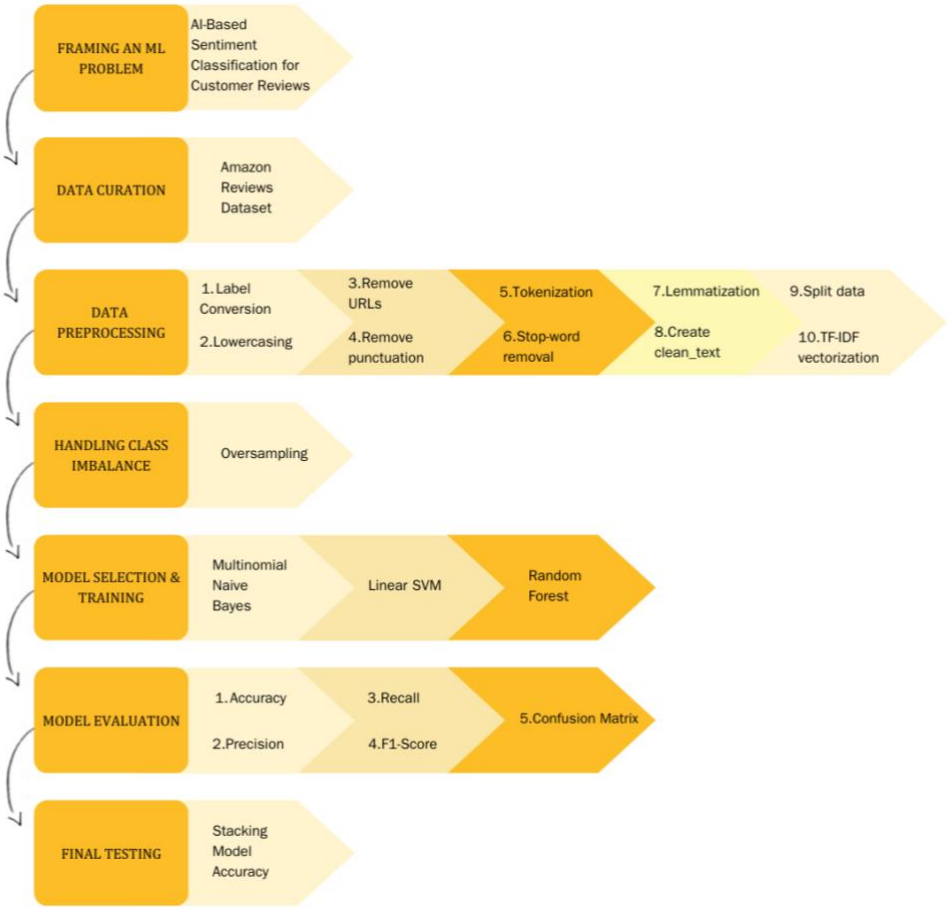

Figure 10: Workflow Diagram of the Proposed ML Module

1) **Framing ML Problem:**

   AI-Based Sentiment Classification for Customer Reviews.

2) **Data Curation:**

   Amazon Reviews Dataset.

3) **Data Preprocessing:**

   Label conversion, lowercasing, removal of URLs and punctuation, tokenization, stop-word removal, lemmatization, creation of clean text, and TF-IDF vectorization.

4) **Handling Class Imbalance:**

   Oversampling the training data.

5) **Model Selection & Training:**
   Train Multinomial Naive Bayes, Linear SVC, and Random Forest base models.
6) **Model Evaluation:**
   Assess individual models based on Accuracy, Precision, Recall, F1 Score, and Confusion Matrix.
7) **Final Testing:**
   Generate the final prediction using the Stacking Ensemble Meta-Learner and report its accuracy.

4.4 Integration with the Larger System

The Sentiment Classification module operates as a standalone component within the platform, but it is tightly linked to the review-handling workflow. When a new review is submitted, the main application passes the raw text directly to this module. The module handles all necessary preprocessing, applies the trained model, and produces a final sentiment label.

Once the prediction is generated, it is returned to the core system and stored with the review. This turns each piece of unstructured text into a clear, searchable sentiment category. As a result, the platform can present reviews in a more organized way, support easier filtering, and give administrators a clearer understanding of overall user feedback trends. By isolating the sentiment logic in its own module, the system remains easier to maintain, update, and extend in the future.

4.5 Expected Benefits

1) **Enhanced Prediction Robustness and Accuracy:**

The stacking ensemble provides a more stable and balanced prediction across classes. It achieved a validation accuracy of 0.8263, combining the strengths of different models to improve overall robustness.

2) Improved Recall for Critical Minority Classes:

Because the data is heavily skewed toward positive reviews, the oversampling and ensemble approach help the model recognize neutral and negative reviews more effectively. This results in better recall for the classes that matter most for identifying issues.

3) Consistent and Scalable Real-time Analysis:

The module can process a large number of reviews efficiently and with consistent quality. This ensures that feedback is analyzed the same way every time.

4) Data-Driven Insights for Strategic Decision Making

Because the system stores sentiment labels rather than just raw text, administrators can easily track satisfaction trends, compare performance across time periods, and identify recurring themes based on sentiment shifts.

5) Extensible Foundation for Future ML Features:

The module provides structured outputs that can be expanded later for example, integrating dashboards, automated alerts, or more advanced analytics. This makes the system more adaptable in the long term.

5 Conclusion

The proposed Machine Learning Model successfully implemented a robust **Stacking Ensemble Classifier** for sentiment categorization in customer review, achieving a **Validation Accuracy of 0.8263**. This development provides a scalable, automated solution that directly addresses the initial problem of inefficient manual review analysis. The model's primary achievement was its effective handling of the severe **class imbalanced** problem through **Random Oversampling**, which notably improved the model's ability to learn the characteristics of minority classes. Furthermore, the ensemble demonstrated high confidence in identifying extreme sentiments, with a Recall of 95% for positive reviews and 88% for negative reviews, ensuring reliable detection of customer satisfaction and dissatisfaction.

The main findings of this study highlights the persistent challenge of classifying **neutral** sentiment. Despite the ensemble approach, the neutral class remained the weakest area of performance, achieving a low Recall of 26%. The implication of this finding is that future research must pivot towards more sophisticated techniques specifically designed to capture subtle linguistic nuances, such as **Aspect-Based Sentiment Analysis (ABSA)** or employing fine-tuned **Transformer-based models**. These methods could potentially offer the contextual understanding required to accurately distinguish ambiguous or mixed-tone reviews from true neutral feedback, thereby further enhancing the system's accuracy and providing richer, more actionable business intelligence.

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The Gemini was utilized as an auxiliary research and consultation tool during the report development. Its support focused on:

Conceptual Alignment: Aiding the team in understanding how researched concepts relate to our specific project design and objectives.

Documentation Refinement: Assisting in drafting descriptive comments and improving the stylistic presentation of the implemented methodology.

Procedural Validation: Consulting on data analysis strategies, specifically the testing procedures across validation and testing sets.